

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).

Assessment Tool Weightage: 25%

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QUESTIONS: 10 Total Marks: 50

Annexure A

Data Interpretation Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent. (5 Marks)
2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. (5 Marks)
3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. (5 Marks)
4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. (5 Marks)
5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. (5 Marks)
6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. (5 Marks)
7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario. (5 Marks)
8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. (5 Marks)
9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward

with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.
(5 Marks)

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.
(5 Marks)

Code of Conduct:

I K.V.S.S. Jeswanth Babu (192472012) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria		Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

ASSESSMENT TOOL 3: COMPREHENSIVE QUESTION ANSWERS

1. Question 1: Gradual Reward Increase (10 → 45)

Data Interpretation Context: The agent's reward values are recorded across multiple training episodes and show a gradual, steady climb from 10 up to 45, without sharp drops or erratic reversals in the overall pattern.

Interpretation & Analysis (Question 1)

- **Interpretation of the Trend:** A consistent, gradual upward trend in cumulative reward is the classic signature of successful reinforcement learning. It shows that the agent is progressively updating its Q-values or policy parameters in a direction that yields better returns, rather than acting randomly.
- **Learning Performance:** The smooth increase (rather than a plateau near 10 or wild oscillation) indicates that the agent's learning algorithm is converging in a stable manner — the value function estimates are becoming more accurate and the temporal-difference error is shrinking episode over episode.
- **Policy Improvement:** Under the Policy Improvement Theorem, a policy that is greedy with respect to an improved value function is guaranteed to be at least as good as the previous policy. The rising reward curve is empirical evidence that this improvement step is being applied correctly at each iteration, moving the agent closer to an optimal policy π^* .
- **Application of Concept:** This pattern is consistent with Generalized Policy Iteration (GPI), where alternating cycles of policy evaluation and policy improvement drive the value estimates and policy toward convergence.
- **Conclusion:** The reward growth from 10 to 45 defends the claim that the agent is learning effectively: it is exploring sufficiently to discover better actions and exploiting that knowledge to steadily raise performance, indicating meaningful policy improvement rather than random fluctuation.

2. Question 2: Epsilon-Greedy — Exploration vs Exploitation

Data Interpretation Context: An epsilon-greedy agent is evaluated at three exploration rates — $\epsilon = 0.1$, 0.3 and 0.5 — and cumulative reward is found to be highest at $\epsilon = 0.1$ and lowest at $\epsilon = 0.5$.

Interpretation & Analysis (Question 2)

- **Exploration vs Exploitation Trade-off:** In an epsilon-greedy policy, the agent exploits its current best-known action with probability $(1 - \epsilon)$ and explores a random action with probability ϵ . A larger ϵ forces more frequent random exploration, while a smaller ϵ allows the agent to exploit learned knowledge more often.
- **Why Reward Falls as Epsilon Rises:** At $\epsilon = 0.5$, the agent selects a random (potentially sub-optimal) action half the time, which repeatedly interrupts exploitation of the best-known action and lowers cumulative reward. At $\epsilon = 0.1$, random exploration occurs only 10% of the time, so the agent exploits its learned optimal action 90% of the time, maximizing cumulative return once a good policy is found.
- **Justification:** Excess exploration is costly once the environment has been sufficiently sampled, because it wastes trials on actions already known to be inferior. Some exploration is still necessary to avoid getting stuck in a locally optimal action, but the marginal benefit of exploration diminishes as knowledge of the environment improves.
- **Application of Concept:** This matches the theoretical exploration–exploitation dilemma in multi-armed bandit and RL settings: too little exploration risks convergence to a sub-optimal action, while too much exploration sacrifices exploitable reward — $\epsilon = 0.1$ sits closer to the balance point for this task.
- **Conclusion:** $\epsilon = 0.1$ provides the better performance because it preserves enough exploration to discover good actions while allowing the agent to exploit that knowledge for the majority of decisions, maximizing cumulative reward.

3. Question 3: Q-Learning vs SARSA Convergence

Data Interpretation Context: Q-Learning and SARSA are compared using average rewards recorded at 100, 200 and 300 episodes, with Q-Learning consistently scoring slightly higher than SARSA at every checkpoint.

Interpretation & Analysis (Question 3)

- **Algorithmic Difference:** Q-Learning is an off-policy algorithm — it updates its Q-values using the maximum estimated future reward ($\max_a Q(s',a)$) regardless of the action actually taken next. SARSA is on-policy — it updates using the reward of the action actually selected by the current (exploratory) policy.
- **Why Q-Learning Scores Higher:** Because Q-Learning always bootstraps toward the greedy (best) next action, it learns the values of the optimal policy directly, which tends to produce higher estimated and realized rewards,

especially once exploration noise decreases. SARSA's on-policy updates incorporate the exploratory ϵ -greedy behaviour into its value estimates, making it more conservative and slightly more cautious near risky states, which can slightly reduce average reward.

- **Convergence Speed:** Since the gap between the two algorithms is consistent from 100 through 300 episodes rather than widening or narrowing sharply, both are converging, but Q-Learning's consistently higher reward at every checkpoint indicates it is converging to a higher-value (closer to optimal) policy faster than SARSA in this setting.
- **Application of Concept:** This is consistent with known theoretical behaviour: off-policy Q-Learning is more aggressive and typically converges faster to the optimal action-value function, while on-policy SARSA converges to a safer policy that accounts for exploration risk.
- **Conclusion:** Q-Learning converges faster and achieves better average performance in this comparison, because its off-policy, max-based updates drive it more directly toward the optimal policy than SARSA's on-policy updates.

4. Question 4: Discount Factor (γ) and Long-Term Reward

Data Interpretation Context: The agent is trained with discount factor values of 0.2, 0.5 and 0.9, and average reward is found to increase steadily as the discount factor increases.

Interpretation & Analysis (Question 4)

- **Role of the Discount Factor:** The discount factor γ in the return $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$ controls how much weight future rewards receive relative to immediate rewards. A low γ (0.2) makes the agent myopic, focusing almost entirely on immediate reward, while a high γ (0.9) makes the agent far-sighted, valuing long-term cumulative reward.
 - **Why Reward Increases with γ :** As γ increases, the agent's value function accounts for the benefit of future states, encouraging it to choose actions that may sacrifice small immediate reward for larger long-term gain. This produces higher average reward over an episode because the policy is optimized for the full trajectory, not just the next step.
- **Justification for $\gamma = 0.9$:** In tasks where the goal or major reward is reached only after a sequence of actions (e.g., reaching a distant goal state), a high discount factor is essential — a low γ would cause the agent to undervalue the eventual payoff and behave sub-optimally. The observed data (highest reward at 0.9) confirms this task rewards long-horizon planning.
- **Application of Concept:** This aligns with Bellman's optimality equation, where the value of a state depends recursively on future discounted values; higher γ propagates reward information further backward through the state space, improving the accuracy of value estimates for long-term planning.
- **Conclusion:** $\gamma = 0.9$ is the most suitable discount factor here, as it allows the agent to properly account for delayed rewards and long-term consequences, resulting in better overall learning performance than the more myopic settings.

5. Question 5: Success Rate Improvement (45% $\rightarrow$ 92%)

Data Interpretation Context: The agent's task success rate is tracked over training and rises from 45% at 50 episodes to 92% at 200 episodes.

Interpretation & Analysis (Question 5)

- **Interpretation of the Improvement:** A steep rise in success rate over training episodes indicates that the agent's policy is becoming substantially more reliable at reaching the goal or completing the task, as it accumulates experience and refines its value estimates.
- **Analysis of the Trend:** The magnitude of improvement (47 percentage points across 150 additional episodes) shows rapid early learning, which is typical of RL agents as they move from near-random initial behaviour to a well informed policy. This is consistent with a learning curve that is still climbing rather than fully flattening.
- **Has an Optimal Policy Been Learned?:** A 92% success rate is strong evidence of substantial policy improvement, but it is not by itself proof of a fully optimal policy, since 8% of trials still fail. An optimal policy would typically be associated with a success rate that has plateaued (stopped increasing) at or near its maximum achievable value,

ideally close to 100% for a deterministic, solvable task.

- **Application of Concept:** Under Generalized Policy Iteration, success rate should continue to rise and then stabilize once the policy converges to π^* ; since the data only shows two points on an upward trend, the agent is still likely in the improvement phase rather than at convergence.
- **Conclusion:** The data supports strong policy improvement and effective learning, but further episodes and a plateau in success rate would be needed to confirm that the agent has converged to a truly optimal policy.